TABLE OF CONTENTS

1 - OVERVIEW

This section applies to systems installed with the 1.5T SRF Cabinets (SRF Cabinet does not contain the PDU or gradient hardware).

Use this procedure to ensure that the power monitors are fully functional, including redundant trip capability. The tests described use a special PSD (pulse sequence data base) to provide excitation for the RF amplifier. The PSD allows the service engineer to selectively force peak power, pulse width, and duty cycle beyond allowable limits. Select set points for testing by modifying Control Variables that override the values normally calculated and downloaded by the PSD.

Power Monitor firmware operation is changed to bypass/test mode by moving "monitor" switches (A and B) on the front of the SSM. This action permits set-point verification without tripping the RF amplifier, thereby eliminating the time consuming necessity of resetting the amplifier after each trip. It is necessary to measure the RF output power and know at what level each monitor tripped.

Three methods are available and listed in the order of preference for measuring head and body RF output power:

- **RF Power Measurement Kit** - Easy to use, preferred method of power measurement.
- **Bird wattmeter** - Must account for cable and load loss to determine actual power.
- **Oscilloscope** - Must know the scope channel correction factor if the scope bandwidth is < 300 MHz. Must account for cable and dummy load loss and the scope bandwidth limitation (channel correction factor) to accurately calculate the power from the observed peak voltage. This method works but the process used to derive the power is somewhat complex and lacks the accuracy of the previous two methods. It is, for this reason, not recommended.

It is *strongly recommended* that the RF Power Measurement Kit be used to obtain an accurate measurement of the RF output power. If it is not possible to obtain an RF Power Measurement Kit then one of the two other methods can be used provided that:

- The Dummy Load and Cables Calibration procedure in Appendix E has been performed and the individual loss values are known.

- The oscilloscope input channel correction factor is known when the scope bandwidth is < 300 MHz *and* the wattmeter is not being used. See Appendix D.

WARNING!

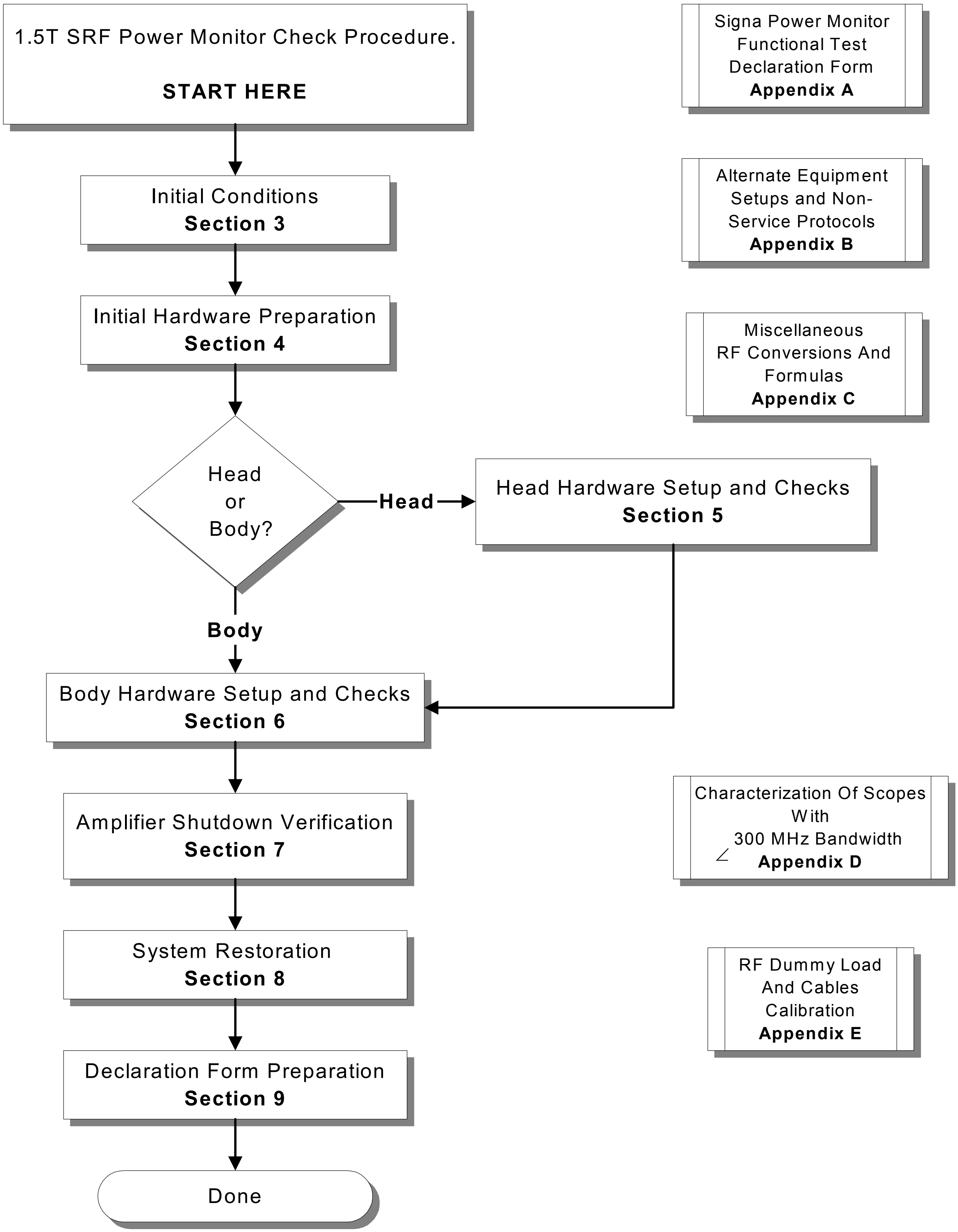
FAILURE TO ACCOUNT FOR THE ACTUAL LOSSES IN THE DUMMY LOAD, CABLES, AND OSCILLOSCOPE WHEN NOT USING THE RF POWER MEASUREMENT KIT TO MEASURE RF POWER CAN RESULT IN SIGNIFICANT MEASUREMENT ERRORS.

1 – OVERVIEW (CONTINUED)

After checking the set points, the actual shutdown of the amplifier is verified: once for monitor A, once for monitor B, and once for the HV relay shutdown. The actual shutdown of the amplifier is verified twice for monitor A (to verify both relay and logic shutdowns), and once for monitor B (connected to amplifier via monitor A). This also confirms that the bypass/test "monitor" switches (A and B) are restored prior to returning the system to normal operation.

A Calibration Flowchart has been provided in Illustration 1-1.

1 – OVERVIEW (CONTINUED)



POWER MONITOR CHECK FLOWCHART
ILLUSTRATION 1-1

2 - TOOLS AND INSTRUMENTS REQUIRED

A Laptop and Serial Port I/F cable (vendor part # 510074) will be required if performing the proprietary procedure. Table 2-1 describes what tools are necessary if the RF Power Measurement Kit is NOT used. Table 2-2 describes all the necessary tools needed if the RF Power Measurement Kit is used.

The use of the RF Power Measurement Kit to complete the RF Power Monitor Check is highly recommended.

TABLE 2-1
TOOLS AND INSTRUMENTS REQUIRED WITHOUT RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	RF Test Cables Kit	46-255816G1
2.	50 ohm, 200 Watt, 30 dB attenuator - Bird Model 8322	46-255837P10
3.	Oscilloscope	46-183029P61
4.	Laptop serial cable (optional)	2124497-46
4.	Wattmeter and appropriate elements (optional)	Not supplied

TABLE 2-2
TOOLS AND INSTRUMENTS REQUIRED WITH RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	RF Power Measurement Kit	46-317724G1 or G2
2.	50 ohm, 200 Watt, 30 dB attenuator - Bird Model 8322 **NOTE:** Only required with G1 Kit	46-255837P10
3.	Oscilloscope	46-183029P61
4.	Laptop serial cable (optional)	2124497-46

3 - INITIAL CONDITIONS

Perform the prerequisite calibration procedures in the sequence listed in Table 3-1.

TABLE 3-1
PRE-REQUISITE PROCEDURES

Step	Procedure Title	When Required
1a	Dummy Load & Cables Calibration – Appendix E (Do if NOT using the RF Power Measurement Kit)	Calibration required whenever using the dummy load and cables. This value will be frequency specific and can change over time.
1b	Scope characterization – Appendix D (Do if NOT using the RF Power Measurement Kit)	This should be done for scopes with bandwidths less than 300 MHz.
2	SRF Max Power RF Out Setup and Calibration (RC5SCA1.DOC on Service Methods CDROM)	At initial installation or when the RFI, UCERD Exciter, or RF Amps are repaired or replaced.

CAUTION

PREVENT EQUIPMENT DAMAGE. ENSURE THAT THE RF AMPLIFIER HAS A DUMMY LOAD CONNECTED TO ITS OUTPUT BEFORE INITIATING TESTS. IF NOT, FAULTS MAY OCCUR AND RF AMPLIFIER COMPONENTS COULD BE PERMANENTLY DAMAGED.

3-1 Pre-requisite Status

Pre-requisite — Signa must be fully functional and properly calibrated.

Pre-requisite — Software must be running with the system capable of normal imaging.

4 - INITIAL HARDWARE PREPARATION

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the **DANGER** message on this page.

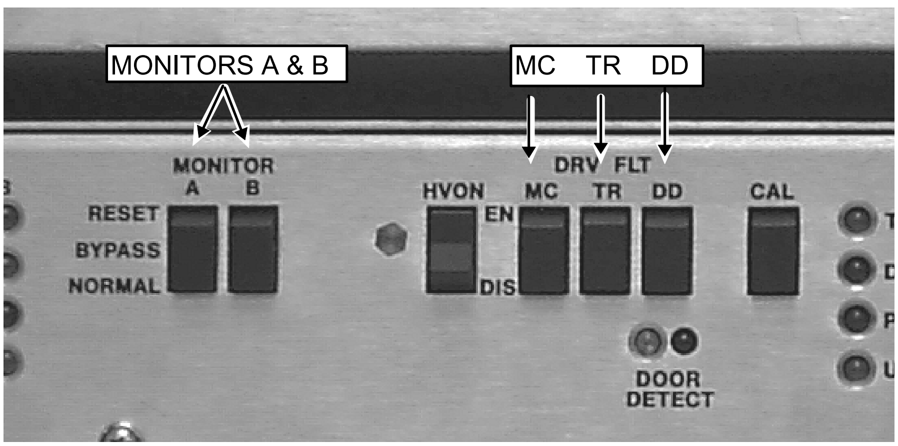
DANGER!!

PROPERTY DAMAGE! TO PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, REMOVE ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL…) FROM THE MAGNET BORE.

2. Remove the front door from the SRF Cabinet.

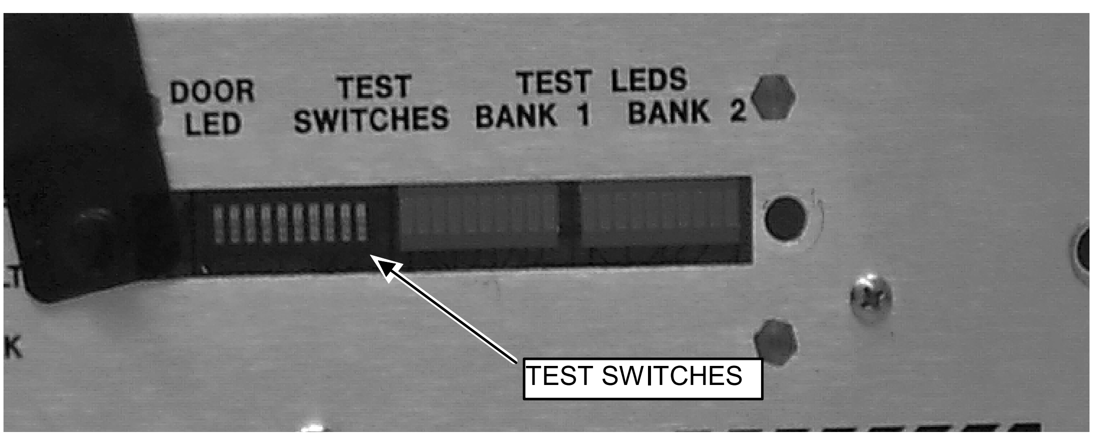
3. See Illustration 4-1. On the front of the SSM place the:

- 2 (two) power MONITOR switches (A and B) to the middle BYPASS position.
- 3 (three) DRV FLT switches to the bottom DIS (disable faults) position.



SSM FRONT PANEL SWITCHES DISABLED
ILLUSTRATION 4-1

4. Remove the test window from the Test Switches and Test LEDS on the front of the SSM. See Illustration 4-2



INSIDE THE TEST WINDOW
ILLUSTRATION 4-2

4 - INITIAL HARDWARE PREPARATION (CONTINUED)

Note

Order of DIP switch placement is important otherwise the system will not recognize changes.

5. Place the first four switches in the order and positions shown in Table 4-1 and leave in these positions for the duration of the test. See Table 4-2 for the alternate proprietary procedure.

TABLE 4-1
SWITCH POSITIONS

SWITCH NUMBER	POSITION
2	UP
3	DOWN
4	UP
1	UP

6. The TEST LED will illuminate. This is the yellow LED to the right of the CAL switch.

4 - INITIAL HARDWARE PREPARATION (CONTINUED)

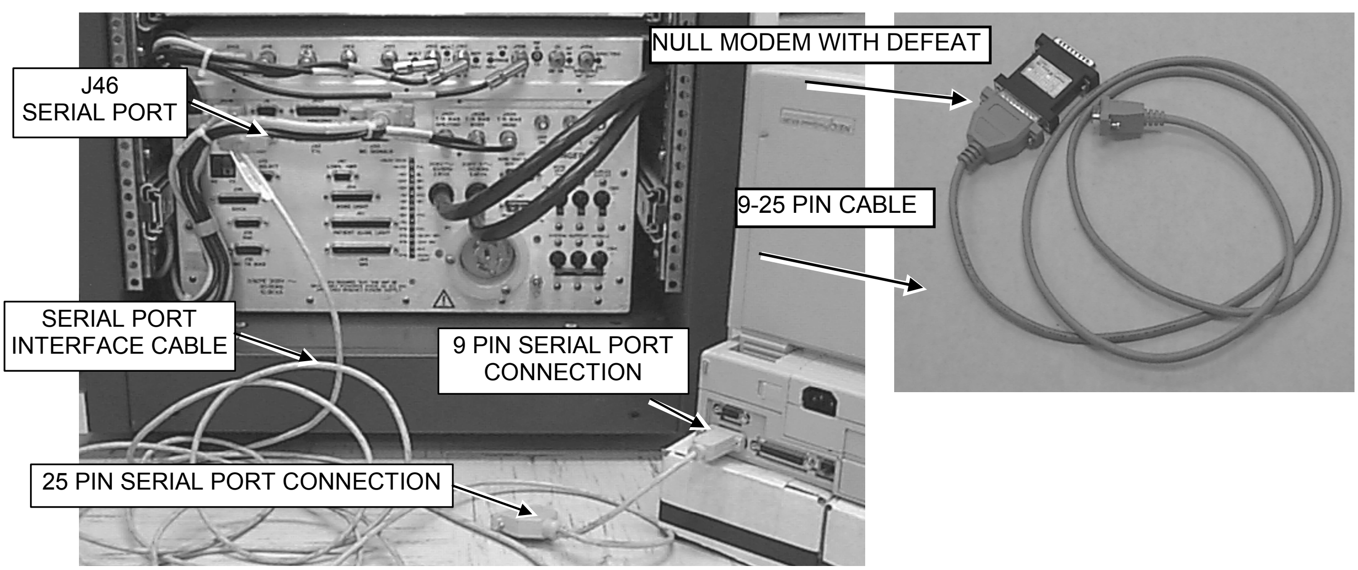
7. Refer to Table 4-2 for the alternate proprietary procedure hardware configuration.

Note

The alternate proprietary procedure utilizes the field laptop and the MONS1.EXE software to detect any faults from RF power monitors A and B. A long (length depends on the distance between the RF/PDU Cabinet and the OW Console at your site) 25 pin female to 25 pin male serial cable (NOT NULL MODEM) can be used with cable 2124497-46 described below in Table 4-2 to remotely monitor and reset power monitor faults from where prescanning changes are being made at the OW Console. While the cable length probably shouldn't exceed 50 feet (15.24 meters) in order for reliable remote monitoring to take place, longer cable lengths (especially if the cable in use is shielded) may work. Alternatively, if the 2124497-6 cable is unavailable, the same locally procured long cable with a null modem adapter affixed to one end and a 25 pin female to 9 pin female adapter connected to the other end could also be used. See Table 4-2 below. Refer to **SECTION 3 - MONS1.EXE Revision 2.2** in **LAPTOP CONFIG AND TROUBLESHOOTING SOFTWARE** (RC3TSC5.DOC) on the 8.X Service Methods CD-ROM for more detailed information concerning the use of MONS1.EXE software.

TABLE 4-2
ALTERNATE PROPRIETARY PROCEDURE USING LAPTOP

Connect the laptop serial port to J46 on the rear of the SSM, using cable part number 2124497-46 (Vendor part # 510074, supplied with every cabinet). See Illustration 4-3. If this cable is not available, use a suitable null modem and 9-25 pin (female on both ends) cable adapter.



4-1 Loading MONS1.EXE – Windows 2000

Note

Only revision 2.2 MONS1.EXE is compatible with all MR systems. This revision, is only available on Signa LX Service Methods CDROM, version 12 and later.

ALTERNATE PROPRIETARY PROCEDURE — LAPTOP SOFTWARE

Start the software diagnostic program MONS1.EXE on the laptop.

Note

The MONS1.EXE program, if not loaded on the service laptop, can be loaded from the Signa LX MR Service Methods CD-ROM 2160623-1. Insert the CD-ROM into the drive. Under Windows 2000 use the cursor to select "**Start**" and then "**Run**" and then "**Browse**". From the Browse window, select the CDROM drive from the down arrow to the right of the "Look in:" entry box. Select the **"Erbtec RF Tools"** folder **or "mrtools"** folder/ directory, then highlight the **Setup.exe** file and select "**Open**" at the bottom of the window. Next, select "**OK**" from the bottom of the Run window.

1) Follow the setup program instructions to install the tool onto the laptop.
2) At the C:\Windows> prompt, type **cd \cclass\rftools<ENTER>**

Note

mons1.exe file "Permissions" must be changed for the program to function under windows 2000.

3) In the "rftools" folder, **right click on the mons1.exe** program.
4) Select **properties,** the mons1.exe (Properties window will open).
5) Click on the **Security Tab** (Top of window).
6) Click on the **add** button. ("Select Users or Groups" window will open).
7) **Select Everyone** and **click on the add** button.
8) Click the **OK** button at the bottom of the Select Users or Groups window.
9) Now click on the **OK** button at the bottom of the properties window.
10) Click on **Kermit.exe** (Kermit terminal window opens).
11) At the prompt within the Kermit terminal window type: **push <ENTER>**.
12) At the prompt that appears, (C:/CLASS/RFTOOLS) type: **mons1.exe <ENTER>**.

Note

mons1.exe will run much slower due to the Kermit terminal window. This is normal.

To exit Kermit: Close the Kermit Window by clicking on the $\boxed{X}$, Select, "**End Now**".

4-2 Loading mons1.exe – Windows 95/98

Note

Only version 2.2 of MONS1.EXE is compatible with all MR system types. This is the version currently available from the Signa LX Service Methods CDROM, revision 12 and later.

TABLE 4-3

ALTERNATE PROPRIETARY PROCEDURE — LAPTOP SOFTWARE

Start the software diagnostic program MONS1.EXE on the laptop.

Note

The MONS1.EXE program, if not loaded on the service laptop, can be loaded from the Signa LX MR Service Methods CD-ROM 2160623-1. Insert the CD-ROM into the drive. Under Windows 95/98 use the cursor to select "**Start**" and then "**Run**" and then "**Browse**". From the Browse window, select the CDROM drive from the down arrow to the right of the "Look in:" entry box. Select the **mrtools** directory, then highlight the **Setup.exe** file and select "**Open**" at the bottom of the window. Next, select "**OK**" from the bottom of the Run window.

Follow the setup program instructions to install the tool onto the laptop. The MONS1.EXE must be run in MS-DOS mode and not from a Windows shell to DOS. Once the installation is complete, from Windows 95/98 perform the following steps:

13) **[Start]**, **[Shut Down...]**, **Restart the computer in MS-DOS mode?**, **[Yes]**

Note

It may take up to 1.5 minutes for some Dell laptops to restart in MS-DOS mode.

2) At the C:\Windows> prompt, type **cd \cclass\rftools<ENTER>**
3) At the C:\CCLASS\ERBTEC> prompt type **mons1.exe<ENTER>**

8. Once the **mons1.exe** program has started then select **N** (next screen) and then **T** (test mode) to reach a screen like that shown in Illustration 4-3.

	MONITOR A	MONITOR B
A Fault is	NOT DETECTED	NOT DETECTED
Press 'esc' to return to previous screen,	'C' to Continue after Fault	

'T' TEST MODE SCREEN (NO FAULT)

ILLUSTRATION 4-3

The screen shown in Illustration 4-3 will be used to remotely and individually monitor Power Monitors A and B for faults.

4 - INITIAL HARDWARE PREPARATION (CONTINUED)

When a fault occurs the word **DETECTED** will be displayed below the power monitor that registered the fault. Illustration 4-4 shows a situation where both power monitors registered a fault.

	MONITOR A	MONITOR B
A Fault is	DETECTED	DETECTED
Press ‘esc’ to return to previous screen, ‘C’ to Continue after Fault		

‘T’ TEST MODE SCREEN (WITH FAULT)
ILLUSTRATION 4-4

The **C** key on the laptop permits the FE to reset the status of Monitors A and B displayed on the Test Mode Screen to “Not Detected” after the FE has manually reset the Power Monitor A and B Reset Switches on the front of the SSM.

CAUTION

When appropriate use the RF Power Measurement Kit, Wattmeter, or Oscilloscope to measure the RF output power and validate that the power monitors are faulting or not faulting at the proper RF power levels. While the Main Screen of the MONS1.EXE software also provides a power measurement readout display for each power monitor, the integrity of each power monitor should be validated independently using the RF Power Measurement Kit, Wattmeter, or Oscilloscope.

9. **If you are using the RF Power Measurement Kit** then calibrate the scope by referring to the RF Power Measurement Kit laminated card set.

    a. Look in the upper right corner of each card and find the card labeled **CAL**.

    b. Configure the scope as in the illustration on the card.

    c. Follow the directions on the card to calibrate the scope using the 4 dBm calibrator.

10. **If you are not using the RF Power Measurement Kit** then complete the Dummy Load and Cables Calibration procedure in Appendix E. If using the scope to measure the RF output power then also make sure that the scope correction factor has been determined for the input channel to be used as described in Appendix D.

5 - HEAD HARDWARE SETUP AND CHECKS

Tests in this section allow the service engineer to selectively force head mode peak power, pulse width, and duty cycle beyond allowable limits.

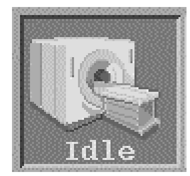
DANGER!!

PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the two **DANGER** messages on this page.

DANGER!!

PERSONAL INJURY! PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RFI BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE.



2. **If using the RF Power Measurement Kit** then refer to the RF Power Measurement Kit laminated card set.

   a. Look in the upper, right corner of each card and find the card labeled **63** (1.5T Head Output).

CAUTION

The head RF output connection is no longer to the non-existent EFB unit, as the reference cards in some of the older kits show, but instead to the J3 output on the rear of the RFI.

   b. Configure the system as shown in the illustration on the card.

   c. Confirm that the rotary attenuator is set to the correct position indicated on the card.

3. **If using the wattmeter or scope** (NOT the RF Power Measurement Kit) to measure power then refer to Appendix B (Alternate Equipment Setup) for the proper system head configuration.

4. At operator workspace, select the scan desktop ICON in the desktop control panel, if you have not already done so.

5 - HEAD HARDWARE SETUP AND CHECKS (CONTINUED)

5. At the operator workspace, prepare the system for a Head Power Monitor scan using the "Service Protocols" procedure in Table 5-1 below. Refer to Appendix B for the non-proprietary protocol.

TABLE 5-1
SCAN PRESCRIPTION - HEAD PM CHECKS

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License. Refer to Appendix B for the non-proprietary protocol. A. **[New Pt]** Id: **geservice <ENTER>** Name: **pmc** Weight (Lb.): **300** Set Patient Protocols to **Service**. B. At front enclosure: Landmark in the Head area—remove all coils. press **LANDMARK**. press **MOVE TO SCAN**. C. At Patient Protocols – select **other**. D. In the protocol field, type **o.23.1<ENTER>** (o=Other, 23.1 =series) to load the head protocol **OR** select **[o.23] [Series 1] [Accept]**. **[OK]** (if required). E. **[Save Series]**. F. **[Prepare to Scan]**. G. **[Research Operations].** H. **[Setup Params]**. Set **TG** to **50**. **[Done]**. I. **[Research Operations]**. **J. [Download]**

6. Refer to Table 5-2 on pages 14 - 16. Note that the table consists of 5 columns (**Test**, **Scan Conditions**, **Adjustments**, **Verify**, **Conclusion**) and 10 individual rows of various head power tests.

Note

It may be advantageous at this point to print a hard copy of Table 5-2. This will provide a quick reference and also a place for making notations.

7. Start at the first **Test** row in Table 5-2. Under the **Scan Conditions** column highlight or type-in the CVs into the CV Name box and then enter the listed corresponding number into the Current Value box. Right mouse-click on the remaining items listed in the column.

8. Advance to the **Adjustments** column on the right and perform the directions listed in the column. Refer to the following steps for measuring the RF output power.

5 - HEAD HARDWARE SETUP AND CHECKS (CONTINUED)

9. **If using the RF Power Measurement Kit** then refer to card **63** (1.5T Head Output) and note the Attenuation value listed on the calibration sticker affixed to the bottom, right of the card.

   a. Measure the amplitude of the RF waveform from the scope face in divisions.

   b. Mouse click on the **ToolBelt Icon** and then **[Utilities], [RF Calculator], [Start].**

   c. Use the RF Calculator program to determine the power output from the amplifier. Enter the actual reference and measured values in place of the sample values in the example below.

(Q)uick RF Calculator
(R)F Calculator

Enter selection: (Q,R) [Q] : **Q**

Quick RF Power Calculator
(A)mplitude from power
(P)ower from amplitude
(Q)uit

Enter selection: (A,P) [P] : **P**

Convert from Scope Amplitude to Power

Oscilloscope calibrated (with tool) to .: 4.00 (dBm)
with a reference amplitude of ....: 8.00 (divisions)
Enter measured amplitude (divisions)....: (0.0..8.00) [7.50] : 6 **←Enter value measured from scope face here.**
Measured power at scope is.............: 1.50 (dBm)

Refer to the RF Power Measurement Kit Quick Reference Card

Enter total component attenuation (dBm): (0.0..75.00) [68.56] : 59.49 **←Enter Attenuation value from the calibration sticker as a <u>POSITIVE</u> number. Omit the negative sign.**

Measured power at source is............: <u>60.99</u> **(dBm) ←Power in dBm at amplifier output**
<u>1256.38</u> **(Watts) ←Power in Watts at amplifier output**

Press ENTER to continue [ ] : **<Enter>**

Quick RF Power Calculator
(A)mplitude from power
(P)ower from amplitude
(Q)uit

Enter selection: (A,P) [P] : **Q**

5 - HEAD HARDWARE SETUP AND CHECKS (CONTINUED)

d. Note the amplifier output power calculated by the RF Calculator program. Continue with the Table instructions.

10. **If using the wattmeter procedure:** Read the wattmeter display and use the formula below to calculate the head RF power. Note that the cable loss factor was determined from the procedure in Appendix E. Record this calculated power and then continue with the Table instructions.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** cable loss factor

11. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit):** Read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator Tool (located at E:\rf\power\pwrcalc.htm on the Service Methods CD-ROM) to calculate the head RF power. Note that the cable loss factor was determined from the procedure in Appendix E and the scope correction factor was determined in Appendix D. Record this calculated power and then continue with the Table instructions.

RF Power Measurement (in watts) Using Oscilloscope And Formula:
$\frac{\left(\frac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

12. When all measurements have been completed then reference the specified limits in the **Verify** column and record the PASS/FAIL status of the test.

13. Perform the steps in **Conclusion** column and then proceed to the next row and repeat the process again. Continue this until all the tests listed in the Table have been completed.

5 - HEAD HARDWARE SETUP AND CHECKS (CONTINUED)

TABLE 5-2
HEAD CHECKS (1.5T) LX

Test	Scan Conditions	Adjustments	Verify	Conclusion
HEAD PEAK POWER *HIGH*	**[Research Operations]** **[Display CVs]** calmode = **2** trig = **7** aset = **120** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure and note Fault 1 power below. **Fault 1 ____________** Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure and note Fault 2 power below. **Fault 2 ____________**	Power (**Fault 1, Fault 2**) is within specifications: Min: 1318 W or 61.2 dBm Max: 1784 W or 62.5 dBm Nom: 1551 W or 61.9 dBm ❑ **Pass** ❑ **Fail**	Set TG to 0. **[Done].** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.
HEAD PEAK POWER *LOW*	**[Research Operations]** **[Display CVs]** aset = **30** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure and note Fault 1 power below. **Fault 1 ____________** Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure and note Fault 2 power below. **Fault 2 ____________**	Power (**Fault 1, Fault 2**) is within specifications: Min: 330 W or 55.2 dBm Max: 446 W or 56.5 dBm Nom: 388 W or 55.9 dBm ❑ **Pass** ❑ **Fail**	Set TG to 0. **[Done].** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.

5 - HEAD HARDWARE SETUP AND CHECKS (CONTINUED)

TABLE 5-2
HEAD CHECKS (1.5T) LX (CONTINUED)

HEAD PULSE WIDTH (PW) *HIGH* (min Limit)	**[Research Operations]** **[Display CVs]** calmode = **1** p1 = **4750** aset = **255** pwset = **100** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Increase TG (from 0) until power measures between: Minimum 200W (53.0 dBm) and Maximum 300W (54.8 dBm).	• SENSE LEDs on SSM are ON • HEAD LED on RFI is ON • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
HEAD PW *HIGH* (Max Limit)	**[Research Operations]** **[Display CVs]** p1 = **5250** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.
HEADPW *LOW* (Min Limit)	**[Research Operations]** **[Display CVs]** p1 = **400** pwset = **10** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Do not change TG.	• SENSE LEDs on SSM are ON) • HEAD LED on RFI is ON • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
HEAD PW *LOW* (Max Limit)	**[Research Operations]** **[Display CVs]** p1 = **600** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.

Note 1

Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.

5 - HEAD HARDWARE SETUP AND CHECKS (CONTINUED)

TABLE 5-2
HEAD CHECKS (1.5T) LX (CONTINUED)

HEAD DUTY CYCLE (DC) HIGH CYCLE (DC) (Min Limit)	**[Research Operations]** **[Display CVs]** t3 = **33333** TR_SLOP = **0** calmode = **3** p3 = **3900** pwset = **255** dcset = **130** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Do not change TG.	• SENSE LEDs on SSM are ON • HEAD LED on RFI is ON • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
HEAD DC *High* (Max limit)	**[Research Operations]** **[Display CVs]** p3 = **4767** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.
HEAD DC *LOW* (Min Limit)	**[Research Operations]** **[Display CVs]** p3 = **750** dcset = **25** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**	Do not change TG.	• SENSE LEDs on SSM are ON • HEAD LED on RFI is ON • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
HEAD DC *LOW* (Max Limit)	**[Research Operations]** **[Display CVs]** p3 =**917** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan]**]	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **NORMAL**. On laptop, press 'C' to continue.

6 - BODY HARDWARE SETUP AND CHECKS

Tests in this section allow the service engineer to selectively force body mode peak power, pulse width, and duty cycle beyond allowable limits.

DANGER!!

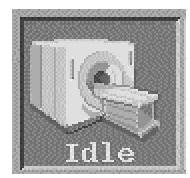
PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Confirm that all the steps in **Section 4 - Initial Hardware Preparation** - have been done.

2. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the two **DANGER** messages on this page.

DANGER!!

PERSONAL INJURY! PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RFI BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE “IDLE” MESSAGE.



3. **If using the RF Power Measurement Kit** then refer to the RF Power Measurement Kit laminated card set.

   a. Look in the upper, right corner of each card and find the card labeled **72** (1.5T Body Output).

CAUTION

The body RF output connection is no longer to the non-existent EFB unit, as the older reference cards from the kit show, but instead to the J4 output on the rear of the RFI. An RF adapter is provided in the RF Power Measurement Kit to connect between the HN J4 body RF output and the RF Power Measurement Kit 40 dB N-connector coupler.

   b. Configure the system as in the illustration on the card.

   c. Confirm that the rotary attenuator is set to the correct position indicated on the card.

4. **If using the wattmeter or scope** (NOT the RF Power Measurement Kit) to measure power then refer to Appendix B (Alternate Equipment Setup) for the proper system body configuration.

5. At operator workspace, select the scan desktop ICON in the desktop control panel, if you have not already done so.

6 - BODY HARDWARE SETUP AND CHECKS (CONTINUED)

6. At the operator workspace, prepare the system for a Body Power Monitor scan using the "Service Protocols" procedure in Table 6-1 below. Refer to Appendix B for the non-proprietary protocol.

TABLE 6-1
SCAN PRESCRIPTION - BODY PM CHECKS

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License. Refer to Appendix B for the non-proprietary protocol. A. **[New Series]** At Patient Protocols – select **other**. B. In the protocol field, type **o.23.2<ENTER>** (o=Other, 23.2 =series) to load the body protocol **OR** select **[o.23] [Series 2] [Accept]**. **[OK]** (if required). C. **[Save Series]**. D. **[Prepare to Scan]**. E. **[Research Operations]**. F. **[Setup Params]**. Set **TG** to **50**. **[Done]**. G. **[Research Operations]**. H. **[Display CVs]**. Highlight CV Name and enter the following: CV name: **dcset <ENTER>**, **255 <ENTER>** CV name: **t3 <ENTER>**, **20000 <ENTER>** I. **[Accept]**. **J. [Research Operations]** **K. [Download]**

7. Refer to Table 6-2. Note that the table consists of 5 columns (**Test**, **Scan Conditions**, **Adjustments**, **Verify**, **Conclusion**) and 10 individual rows of various head power tests.

Note

It may be advantageous at this point to print a hard copy of Table 6-2. This will provide a quick reference and also a place for making notations.

8. Start at the first **Test** row in Table 6-2. Under the **Scan Conditions** column highlight or type-in the CVs into the CV Name box and then enter the listed corresponding number into the Current Value box. Right mouse-click on the remaining items listed in the column.

9. Advance to the **Adjustments** column on the right and perform the directions listed in the column. Refer to the following steps for measuring the RF output power.

6 - BODY HARDWARE SETUP AND CHECKS (CONTINUED)

10. **If using the RF Power Measurement Kit** refer to card **72** (1.5T Body Output) and note the Attenuation value listed on the calibration sticker affixed to the bottom, right of the card.

    a. Measure the amplitude of the RF waveform from the scope face in divisions.

    b. Mouse click on the **ToolBelt Icon** and then **[Utilities], [RF Calculator], [Start].**

    c. Use the RF Calculator program to determine the power output from the amplifier. Enter the actual reference and measured values in place of the sample values in the example below.

Quick RF Power Calculator
(A)mplitude from power
(P)ower from amplitude
(Q)uit

Enter selection: (A,P) [P] : **P**

Convert from Scope Amplitude to Power

Oscilloscope calibrated (with tool) to .: 4.00 (dBm)
with a reference amplitude of ....: 8.00 (divisions)
Enter measured amplitude (divisions)....: (0.0..8.00) [7.50] : 6 **←Enter value measured from scope face here.**
Measured power at scope is..............: 1.50 (dBm)

Refer to the RF Power Measurement Kit Quick Reference Card

Enter total component attenuation (dBm): (0.0..75.00) [68.56] : 59.49 **←Enter Attenuation value from the calibration sticker as a <u>POSITIVE</u> number. Omit the negative sign.**

Measured power at source is............: <u>60.99</u> **(dBm) ←Power in dBm at amplifier output**
<u>1256.38</u> **(Watts) ←Power in Watts at amplifier output**

Press ENTER to continue [ ] : **<Enter>**

Quick RF Power Calculator
(A)mplitude from power
(P)ower from amplitude
(Q)uit

Enter selection: (A,P) [P] : **Q**

    d. Note the amplifier output power calculated by the RF Calculator program. Continue with the Table instructions.

6 - BODY HARDWARE SETUP AND CHECKS (CONTINUED)

11. **If using the wattmeter procedure:** Read the wattmeter display and use the formula below to calculate the body RF power. Note that the dummy load and cable loss factor was determined from the procedure in Appendix E. Record this calculated power and then continue with the Table instructions.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** dummy load and cable loss factor

12. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit):** Read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator Tool (located at E:\rf\power\pwrcalc.htm on the Service Methods CD-ROM) to calculate the body RF power. Note that the dummy load and cable loss factor was determined from the procedure in Appendix E and the scope correction factor was determined in Appendix D. Record this calculated power and then continue with the Table instructions.

RF Power Measurement (in watts) Using Oscilloscope And Formula:
$\dfrac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

13. When all measurements have been completed then reference the specified limits in the **Verify** column and record the PASS/FAIL status of the test.

14. Perform the steps in **Conclusion** column and then proceed to the next row and repeat the process again. Continue this until all the tests listed in the Table have been completed.

6 - BODY HARDWARE SETUP AND CHECKS (CONTINUED)

TABLE 6-2
BODY CHECKS (1.5T)

TEST	SCAN CONDITIONS	ADJUSTMENTS	VERIFY	CONCLUSION
BODY PEAK POWER *HIGH*	**[Research Operations]** **[Display CVs]** calmode = **2** trig = **7** aset = **120** **[Accept]** **[Research Operations]** **[Download]** **[Manual Prescan].**	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure and note Fault 1 power below. **Fault 1** ____________ Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure and note Fault 2 power below. **Fault 2** ____________	Power (**Fault 1, Fault 2**) is within specifications: Min: 8793.9W, 69.44dBm Max: 11897.7W, 70.75dBm Nom: 10345.8W, 70.15dBm ❑ **Pass** ❑ **Fail**	Set TG to 0. **[Done].** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.
BODY PEAK POWER *LOW*	**[Research Operations]** **[Display CVs]** aset = **30** **[Accept]** **[Research Operations]** **[Download]** **[Manual Prescan].**	Increase Transmit Gain (TG) until laptop or SSM LED indicates fault occurred on one of the monitors; measure and note Fault 1 power below. **Fault 1** ____________ Continue increasing TG until laptop or SSM LED indicates other monitor has faulted; measure and note Fault 2 power below. **Fault 2** ____________	Power (**Fault 1, Fault 2**) is within specifications: Min: 2198.5W or 63.42dBm Max: 2974.5W or 64.73dBm Nom: 2586.5W or 64.13dBm ❑ **Pass** ❑ **Fail**	Set TG to 0. **[Done].** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.

6 - BODY HARDWARE SETUP AND CHECKS (CONTINUED)

TABLE 6-2
BODY CHECKS (1.5T) (CONTINUED)

TEST	SCAN CONDITIONS	ADJUSTMENTS	VERIFY	CONCLUSION
BODY PULSE WIDTH(PW) *HIGH* (Min Limit)	**[Research Operations]** **[Display CVs]:** aset = **255** calmode = **1** pwset = **100** p1 = **4750** **[Accept]** **[Research Operations]** **[Download]** **[Manual Prescan].**	Increase TG (from 0) until power measures between: Minimum 2 kW (63.01 dBm) and Maximum 3 kW (64.77 dBm).	• SENSE LEDs on SSM are ON, • BODY LED on RFI is ON, • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
BODY PW *HIGH* (Max Limit)	**[Research Operations]** **[Display CVs]:** p1 = **5250** **[Accept]** **[Research Operations]** **[Download]** **[Manual Prescan].**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.
BODY PW *LOW* (Min Limit)	**[Research Operations]** **[Display CVs]:** p1 = **400** pwset=**10** **[Accept]** **[Research Operations]** **[Download]** **[Manual Prescan].**	Do not change TG.	• SENSE LEDs on SSM are ON, • BODY LED on RFI is ON, • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
BODY PW *LOW* (Max Limit)	**[Research Operations]** **[Display CVs]:** p1 = **600** **[Accept]** **[Research Operations]** **[Download]** **[Manual Prescan].**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press 'C' to continue.

Note 1

Allow at least 12 seconds after looping starts before checking status of FAULT LEDs.

6 - BODY HARDWARE SETUP AND CHECKS (CONTINUED)

TABLE 6-2
BODY CHECKS (1.5T) (CONTINUED)

TEST	SCAN CONDITIONS	ADJUSTMENTS	VERIFY	CONCLUSION
BODY DUTY CYCLE (DC) *HIGH* (Min Limit)	**[Research Operations]** **[Display CVs]:** t3=**33333** TR_SLOP=**0** calmode = **3** pwset = **255** dcset = **130** p3 = **3900** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan].**	Do not change TG.	• SENSE LEDs on SSM are ON, • BODY LED on RFI is ON, • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
BODY DC *HIGH* (Max Limit)	**[Research Operations]** **[Display CVs]:** p3 = **4767** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan].**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press ’C’ to continue.
BODY DC *LOW* (Min Limit)	**[Research Operations]** **[Display CVs]:** p3 = **750** dcset = **25** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan].**	Do not change TG.	• SENSE LEDs on SSM are ON, • BODY LED on RFI is ON, • FAULT LEDs (both OFF) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done].**
BODY DC *LOW* (Max Limit)	**[Research Operations]** **[Display CVs]:** p3 = **917** **[Accept]** **[Research Operations] [Download]** **[Manual Prescan].**	Do not change TG.	• FAULT LEDs (both ON) (Note 1) ❑ **Pass** ❑ **Fail**	**[Done],** Place both Monitor A and B switches to **RESET**, then **BYPASS**. On laptop, press ’C’ to continue.

7 - AMPLIFIER SHUTDOWN VERIFICATION

Tests in this section verify that each power monitor is capable of shutting down the RF Amplifier independently of the other, and further verify both the logic and relay shutdowns for each channel. Setpoints are not checked in this section — this test is forcing a shutdown.

When one or both of the monitors is in bypass mode, the HV relay shutdown is disabled.

1. Enable power monitor A by placing the monitor A switch on the front panel of the SSM in the bottom Normal (down) position.

2. At the operator workspace, prepare the system for a Amp Shutdown Verification scan using the "Service Protocols" procedure in Table 7-1 below. Refer to the body protocol in Appendix B for the non-proprietary protocol.

Note

Verify that the Head Coil has been removed from the patient table before continuing.

TABLE 7-1

SCAN PRESCRIPTION - HIGH VOLTAGE RELAY SHUTDOWN — BODY MODE

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Series]**
 At Patient Protocols – select **other**.
B. In the protocol field, type **o.23.3<ENTER>** (o=Other, 23.3 =series) to load the body protocol
OR select **[o.23] [Series 3] [Accept]**.
[OK] (if required).
C. **[Save Series]**
D. **[Prepare to Scan]**.
E. **[Research Operations]**.
 [Setup Params]. Set **TG** to **0**. **[Done]**.
F. **[Research Operations]**.
G. **[Display CVs]**. Highlight CV Name and enter the following:
 CV name: **trig <ENTER>**, **1 <ENTER>**
 CV name: **aset <ENTER>**, **120 <ENTER>**
 CV name: **calmode <ENTER>**, **2 <ENTER>**
 CV name: **pwset <ENTER>**, **255 <ENTER>**
 CV name: **dcset <ENTER>**, **255 <ENTER>**
 CV name: **p1 <ENTER>**, **3100 <ENTER>**
 CV name: **p3 <ENTER>**, **2400 <ENTER>**
H. [Accept]
I. **[Research Operations]**
J. **[Download]**.
K. Select **[Manual Prescan]**.

7 - AMPLIFIER SHUTDOWN VERIFICATION (CONTINUED)

CAUTION

The system error log will be checked in the next step for a power monitor fault error message. Make sure the error log Viewing Level selection is set to "ALL" before reading the error log. The error message may not otherwise be visible.

3. Increase TG until monitor A FAULT LED illuminates. Quickly verify the following:
   - The FAULT LED on for monitor A (resets automatically after 1 second).
   - Software Message Log - Software will log a power monitor fault in the message log.
   - Verify oscilloscope or Wattmeter (press FWD PEP then MAX) reads zero RF power.

Note

Do not perform any operations at the keyboard while In Place Error Recovery (IPER) is resetting the RF amplifier.

4. While waiting for the system to recover, place monitor B in normal mode by placing the monitor B switch on the front panel of the SSM in the Normal (down) position. Now both monitors A and B should be in normal mode.
5. After IPER has reset the RF amplifier (the "Please Press Start Scan Button" message appears in the upper left message window), decrease TG by 25 counts (2.5 dB) and then press START SCAN on the operator's console to initiate looping.
6. Increase TG until the FAULT LED for monitor A or B illuminates. Quickly verify the following:
   - FAULT LED on for monitor A or B (resets automatically after two seconds).
   - Software Message Log - Software will log a power monitor fault in the message log.
   - Verify oscilloscope or Wattmeter (press FWD PEP then MAX) reads zero RF power.

Note

Do not perform any operations at the keyboard while In Place Error Recovery (IPER) is resetting the RF amplifier.

7. While waiting for the system to recover, disable power monitor A by placing the monitor A switch on the front panel of the SSM in the Bypass position.
8. After IPER has reset the RF amplifier (the "Please Press Start Scan Button" message appears in the upper left message window), decrease TG by 25 counts (2.5 dB) and then press START SCAN on the operator's console to initiate looping.
9. Increase TG until monitor B FAULT LED illuminates. Quickly verify the following:
   - FAULT LED on for monitor B (resets automatically after two seconds).
   - Software Message Log - Software will log a power monitor fault in the message log.
   - Verify oscilloscope or Wattmeter (press FWD PEP then MAX) reads zero RF power.

7 - AMPLIFIER SHUTDOWN VERIFICATION (CONTINUED)

Note

Do not perform any operations at keyboard while In Place Error Recovery (IPER) is resetting the RF amplifier.

10. After IPER has reset the RF amplifier (the “Please Press Start Scan Button” message appears in the upper left message window), select **[Done]**.

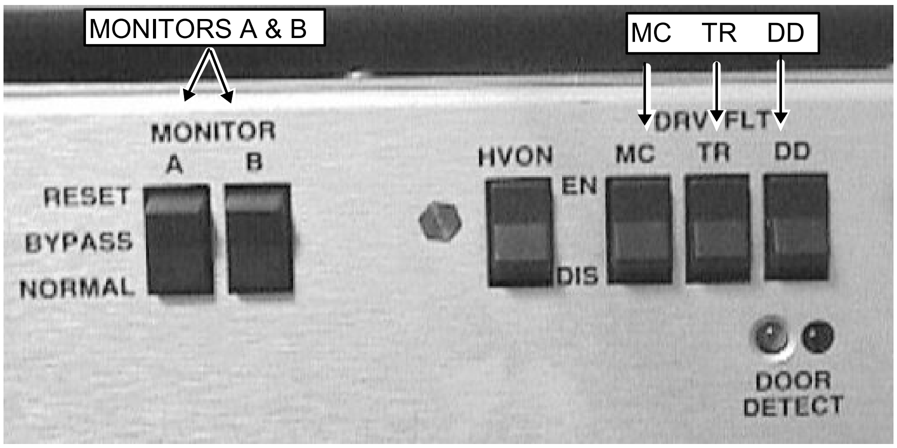
11. In the Rx Manager, select **[End Exam]**, then **[Confirm]**.

12. Proceed to **SECTION 8 – SYSTEM RESTORATION**.

8 - SYSTEM RESTORATION

1. Verify the system is not scanning and the scan desktop icon is displaying the "Idle" message.

2. Refer to Illustration 8-1. At the front of the SSM place the:
   - 2 (two) power MONITOR switches (A and B) to the bottom NORMAL position.
   - 3 (three) DRV FLT switches to the top EN (enable faults) position.



FRONT PANEL SWITCHES ENABLED
ILLUSTRATION 8-1

3. Place DIP switches 4, 3, 2, and 1 of the Test Switches in the DOWN position or disconnect laptop computer from J46 on the rear of the SSM.

   Note
   If using the Laptop perform the following steps to exit MONS1.EXE and return to Windows:
   A) Follow screen directions to return to DOS.
   B) At the DOS C:\CCLASS\RFTOOLS> prompt, type **EXIT<ENTER>**.

4. Ensure that the dummy load (all test equipment) is disconnected from the rear of the cabinet.

5. Ensure that the head and body Heliax cables are reconnected to the rear of the cabinet.

6. Complete one head scan satisfactorily.

7. Complete one body scan satisfactorily.

8. Close the rear door and replace the front cover on the cabinet.

Prepare and file power monitor functional check data sheet according to current company policy. See **SECTION 9 – DECLARATION FORM PREPARATION**.

9 - DECLARATION FORM PREPARATION

1. Prepare Direction 2212504, *Signa Release 8.X SRF 1.0T/1.5T Cabinet Power Monitor Functional Test Declaration Form, Appendix A*. Refer to procedure for Data Sheet for Power Monitor Test Declaration Form.

2. File completed declaration form per current company policy.

APPENDIX A — SIGNA POWER MONITOR FUNCTIONAL TEST DECLARATION FORM

Signa Release 8.X SRF 1.0T/1.5T CABINET
Power Monitor Functional Test Declaration

SITE

(NAME)	(SYSTEM ID, ROOM NUMBER, ETC., IF APPLICABLE)
(STREET)	(SYSTEM CABINET SERIAL NUMBER)
(CITY, STATE, ZIP CODE)	(CURRENT SYSTEM SOFTWARE RELEASE)

TEST EQUIPMENT

(BIRD WATTMETER SERIAL NO.) (OSCILLOSCOPE SERIAL NO.) (RF POWER MEASUREMENT KIT SERIAL NO.)

(DUMMY LOAD PLUS CABLES ATTEN FACTOR) (AMP TO WATTMETER CABLE ATTEN FACTOR)

PERFORMANCE

FUNCTION CHECK STATUS	PASS
BODY PEAK POWER TEST	❑
BODY PULSE WIDTH TEST	❑
BODY DUTY CYCLE TEST	❑
HEAD PEAK POWER TEST	❑
HEAD PULSE WIDTH TEST	❑
HEAD DUTY CYCLE TEST	❑
SHUTDOWN VERIFICATION	❑

CONFIGURATION FILE ENTRIES

	Cable loss coefficient	Coil loss coefficient
Head Coil		
Body Coil		

AMP Cal Head		ABSOLUTE AVG SAR	
AMP Cal Body		ABSOLUTE PEAK SAR	
Average SAR Factor		Peak SAR Factor	

RESTORATION CHECK LIST

- ❑ On the front panel of the SSM, place the two monitor switches in the Normal position, and place the three DRV FLT switches in the enable faults position (EN).
- ❑ Ensure that the dummy load is disconnected from the rear of the cabinet.
- ❑ Ensure that the head and body Heliax cables are reconnected to the rear of the cabinet.
- ❑ Disconnect laptop computer from J46 on the rear of the SSM.
- ❑ Complete one head scan satisfactorily.
- ❑ Complete one body scan satisfactorily.
- ❑ Replace the front door on the cabinet.
- ❑ Prepare and file Direction 2212504, *Signa Horizon 5.X & 8.X SRF 1.0T/1.5T Power Monitor Functional Test Declaration Form* according to company policy.

(PERFORMED BY - PLEASE PRINT)

(SIGNATURE) (DATE)

DIRECTION 2212504 REV 2 (March 16, 1999)

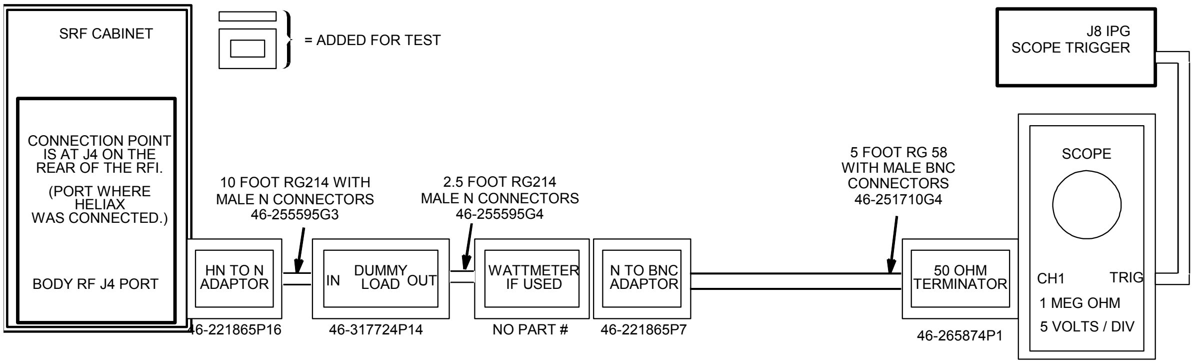
APPENDIX B — ALTERNATE EQUIPMENT SETUPS & NON-SERVICE PROTOCOLS

Use this section if the RF Power Measurement Kit is NOT going to be used. This section contains the setup information and protocols necessary to measure the body and head RF output power using either a wattmeter or oscilloscope. If the wattmeter is not used it is strongly recommended that an oscilloscope with a bandwidth of 300 MHz or greater be used for making the RF output power measurements. A 100 MHz oscilloscope can be used to make these measurements **provided** that the oscilloscope input channel to be used has been properly characterized and the resulting correction factor is known. The oscilloscope must be characterized and the correction factor must be known to account for measurement error at 63.86 MHz error due to bandwidth limitations of the 100 MHz oscilloscope.

TABLE B-1

TOOLS AND INSTRUMENTS REQUIRED WITHOUT RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	RF Test Cables Kit	46-255816G1
2.	50 ohm, 200 Watt, 30 dB attenuator - Bird Model 8322	46-255837P10
3.	Oscilloscope	46-183029P61
4.	Laptop serial cable (optional)	2124497-46
4.	Wattmeter and appropriate elements (optional)	Not supplied



ALTERNATE BODY EQUIPMENT SETUP FOR WATTMETER OR SCOPE MEASUREMENT
ILLUSTRATION B-1

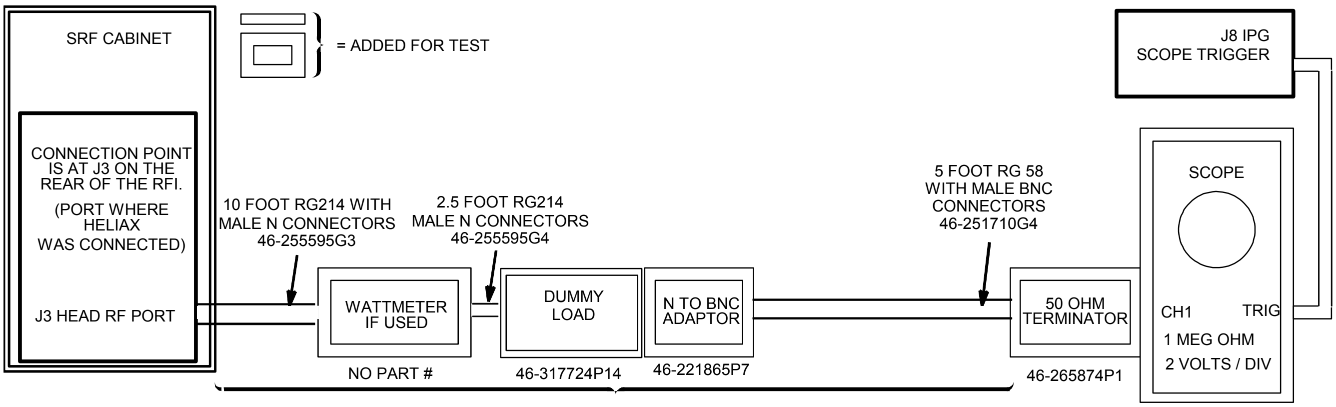
Note
If the wattmeter is not available then bypass it's connection in the circuit using an N-female to N-female connector 46-265875P2.

NON-SERVICE BODY PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**body pmc**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Body] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo] [Accept]**
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[200]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[10]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [R/L]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**
[Research Operations]	**[Setup Params]**
Set TG	**50 [Done]**
[Research Operations]	**[Download]**
[Prepare to Scan]	

If performing Amplifier Shutdown Verification of Section 7 then be sure to enter the CVs per Table 7-1, items F - H.



ALTERNATE HEAD EQUIPMENT SETUP
ILLUSTRATION B-2

Note

If the wattmeter is not available then bypass it's connection in the circuit using an N-female to N-female connector 46-265875P2.

NON- SERVICE HEAD PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**head pmc**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Head] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo]**
	[Accept]
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[200]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[10]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [R/L]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**
[Research Operations]	**[Setup Params]**
Set TG	**50 [Done]**
[Research Operations]	**[Download]**
[Prepare to Scan]	

APPENDIX C — SOFTWARE AND FORMULAE FOR MISCELLANIOUS RF CONVERSIONS

Software are described and formulae are presented in this section that will allow the FE to perform various conversions and calculations.

C-1 RF Conversion Software

C-1-1 RF Power Calculator

The first tool, the RF Power Calculator, is a UNIX-based application available on the LX system under the **Toolbelt icon**, **[Utilities]**, **[RF Calculator], [Start]**. Watts to dBm, dBm to Watts, relative volts to relative power, and relative power to relative volts conversions can be done with the RF Power Calculator tool.

C-1-2 Power Calculator Tool

The second tool, the Power Calculator, is an HTML document that is available from the 8.X Service Methods CD-ROM. This tool will run from either the laptop PC or MR host computer. The tool can manually be started from the laptop PC by selecting **Start, Run,** and then entering the following file pathname in the “Open” entry box: E:\rf\power\pwrcalc.htm. Next, select **OK**. The tool is divided into 3 sections. The first section allows one to calculate RF power from measured peak voltage when using a 100 MHz oscilloscope and a properly characterized RF cables kit and dummy load. The second section provides Vpp to dBm, dBm to Watts, and dBm to Vpp conversion capability. The third section provides short instructions and a means for comparing the attenuation in dB reported on each RF Power Measurement Kit Card with the measured Magnitude Squared Attenuation Factor reported by the Attenuation Test tool that is described in **APPENDIX D – DUMMY LOAD AND CABLES CALIBRATION.**

C-2 RF Conversion Formulae

These are provided to assist the experienced FE with RF power measurement and troubleshooting. The RF conversions below are presented in two different formats. The first format is the actual mathematical formula. The second format assumes that you are entering the data into the LX system calculator exactly as you read it (Access the LX system calculator by right mouse clicking in the background and then selecting the calculator from the Root menu.).

VP-P to dBm Calculation:

$$\text{dBm} = 20 \log \left(\frac{Vpp}{0.632} \right)$$

VP-P **[÷]** 0.632 **[=]** **[LOG]** **[*]** 20 **[=]** dBm.

APPENDIX C — SOFTWARE AND FORMULAE FOR MISCELLANIOUS RF CONVERSIONS (CONTINUED)

dBm to VP-P Calculation:

Vpp = $10^{\left(\frac{dBm}{20}\right)}$ **X** 0.632 or Vpp = antilog $\left(\frac{dBm}{20}\right)$ **X** 0.632

dBm **[÷]** 20 **[=]** **[INV LOG]** **[∗]** 0.632 **[=]** VP-P.

Example: Using 3.65 dBm, the Scope Power reading listed on a particular Kit card, and the LX system calculator to determine the equivalent Vpp:

3.65 **[÷]** 20 **[=]** (0.1825) **[INV LOG]** (1.5222991) **[∗]** 0.632 **[=]** (0.9620931) VP-P.

dBm to Watts Calculation:

Watts = $10^{\left(\frac{dBm}{10}\right)}$ **X** 0.001 or Watts = antilog $\left(\frac{dBm}{10}\right)$ **X** 0.001

dBm **[÷]** 10 **[=]** **[INV LOG]** **[=]** **[∗]** 0.001 **[=]** Total Watts.

C-3 1.5T Division Conversions

The 1.5T division conversion below may be useful for the FE using the RF Power Measurement Kit and attempting to measure a signal that requires more resolution that the "division" unit provides. These examples are provided as a general reference and are not intended to be used to perform the power monitor check.

1.5T Division Conversion example:

- 1.5T sites using the 1.5T Scope Calibrator and 100 MHz Oscilloscope equipment:

$$\frac{\text{1.00 VPP (4.0 dBm)}}{\text{Set to 8 Major Div}} = \frac{\text{0.125 VPP each Major Division}}{\text{5 Minor Divisions}} = \text{0.025 VPP each Minor Division}$$

$$\text{0 dBm} = \frac{\text{0.632 VPP}}{\text{0.025 VPP each Minor Division}} = \textbf{25.28 Minor Divisions}$$

25.28 Minor Divisions = 5 Major Divisions and .28 of a Minor Division

APPENDIX D — CHARACTERIZATION FOR SCOPES WITH < 300 MHZ BANDWIDTH

WARNING!

THIS PROCEDURE REQUIRES THAT THE SCOPE IN USE HAS A BANDWIDTH OF AT LEAST 100 MHZ. DO NOT ATTEMPT TO PERFORM RF POWER MEASUREMENTS WITH AN OSCILLOSCOPE THAT HAS A BANDWIDTH LESS THAN 100 MHZ. SIGNIFICANT MEASUREMENT ERRORS CAN RESULT.

The scope correction factor for the scope channel in use must be known for oscilloscopes with a bandwidth < 300 MHz. One may assume that the correction factor is 1 (unity) for oscilloscopes with a bandwidth ≥ 300 MHz. Do not assume a correction factor for an input channel and do not assume that all the scope input channels have the same correction factor. Each channel must be characterized separately. There are three ways that this can be done.

D-1 Scope Channel Characterization Done During Calibration

One way that the scope channel characterization can be accomplished is to request this information from the metrology lab the next time the scope is sent back for calibration. Attach a tag in a conspicuous location on the scope before it is sent back for calibration requesting this information. Ask that the percent of signal loss at 63.86 MHz (1.5T) and 42 MHz (1.0T) be measured at the 5 volt/div and 2 volt/div settings (for body and head measurements, respectively) for both scope channels and that this information be included with the returned scope. The metrology lab can usually do this upon request and without charge during the calibration process.

D-2 Scope Channel Characterization Done By Comparison

A 400 MHz scope exhibits negligible measurement error at 63.86 MHz or 42 MHz. If a 400 MHz scope is available then it can be compared side-by-side on site with a scope of < 300 MHz bandwidth. Be sure the scope inputs have a 1Mohm impedance and then are terminated with a 50 ohm feed-through adapter before making the measurements. Be sure to characterize all the inputs that are to be used. The signal loss at 63.86 MHz (1.5T) and 42 MHz (1.0T) for each channel at the 5 volt/div and 2 volt/div settings can be determined from the ratio of signal measured on the 100 MHz scope to that seen on the 400 MHz scope

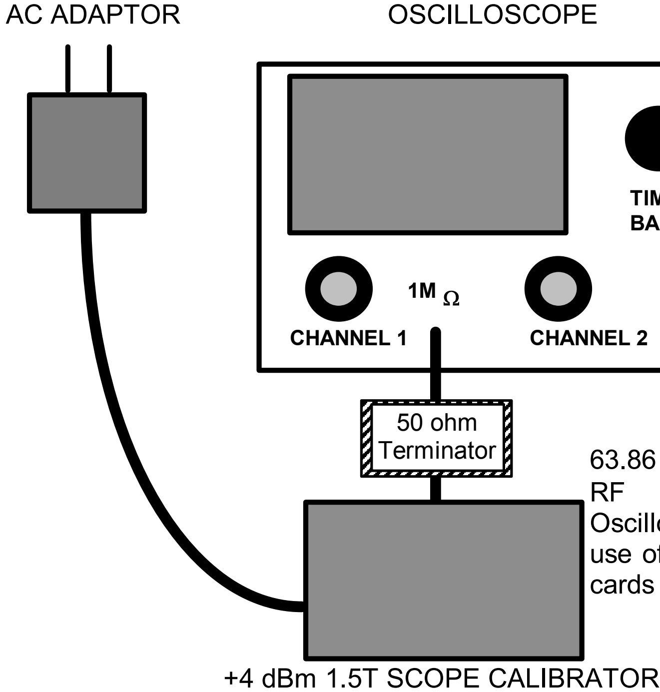
$$\frac{\text{100 MHz scope measured Vpp signal}}{\text{400 MHz scope measured Vpp signal}} = \text{scope correction factor}$$

. As long as the same cables are used the loss can also be determined for each cable.

D-3 Scope Characterization Using the RF Power Measurement Kit Signal Reference

The RF Power Measurement Kit contains a 63.86 MHz, 4dBm (1 Vpp) calibrated signal reference that can be used to characterize the scope input channels. Be aware that these values are accurate only for 1.5T systems.

1. Connect the calibrated reference to the scope as in Illustration D3-1 below.



Note
63.86 MHz oscillator with 4dBm/1.00 VP-P RF Output into 50 Ω termination. Oscilloscopes less than 300 MHz require the use of this 1.5T Scope Calibrator. All 1.5T cards provide Scope Readings in Divisions.

1.5T SCOPE CALIBRATOR CONNECTIONS
ILLUSTRATION D3-1

2. Set the scope amplitude for the channel to be checked to 0.2 volts/div.
3. Set the sweep rate to 5 nsec.
4. Confirm that the channel input impedance is set for 1Mohm and not 50 ohms.
5. Confirm that the scope bandwidth limit switch is not enabled.
6. Confirm that the "Uncal" knob is not rotated and the channel is not uncalibrated.
7. Carefully measure the peak to peak voltage from the calibrated source. The measured value should be 1 Vpp or less.
8. Calculate the correction factor (it should be less than 1):

$$\frac{\text{Vpp measured}}{1\ \text{Vpp}} = \text{correction factor}$$

9. Repeat this process and derive the correction factor for the other scope input channel(s).
10. Note these correction factors for each channel on a piece of tape affixed to the scope for later reference.

APPENDIX E — DUMMY LOAD AND CABLES CALIBRATION

Description - This procedure provides directions for determining the true loss attributable to the dummy load and cables used when measuring the RF output power with a wattmeter or oscilloscope of 100 MHz bandwidth or greater. It is necessary to know and account for the actual loss these components contribute in order to accurately measure RF power. This procedure is not needed if using the RF Power Measurement Kit. The RF Power Measurement Kit has already been calibrated so that this loss is known and accounted for.

Note

This tool will accurately measure attenuation values up to 50dB. The combined attenuation of the dummy load and cables is well below this value. The accuracy of the tool begins to diminish when measuring attenuation values over 50dB.

CAUTION

WHEN MEASURING HIGH FREQUENCY (IN THIS CASE, 42 OR 64 MHZ) ON ANY 100 MHZ SCOPE, THERE MAY BE AN ERROR DUE TO THE BANDWIDTH LIMITATIONS OF THE SCOPE. (IF NECESSARY, READ 'SCOPE TRIVIA' AT THE END OF THIS PROCEDURE.)

E-1 Overview

Test cables long enough to reach the cables, connectors and dummy load to be tested are connected between the Exciter RF Output (System Cabinet J1) and Receiver Body Input (System Cabinet J2). Receiver gains (R1 & R2) and transmit gain (TG) are set for near full scale reading on the power spectrum during prescan calibration. A reference scan is taken and stored in a raw file. The Attenuation Test Tool is used to calculate the baseline factor from the reference scan for the test cables (i.e., there is some loss from the test cables).

The dummy load and/or cable(s) to be tested are next inserted in series with the test cables and another scan is taken. Again, the Attenuation Test Tool is used to determine the "Magnitude Squared Attenuation Factor" (i.e., how much has the test signal been attenuated?). This attenuation factor is used in the RF power calibration process to accurately calculate the RF power level.

Note

If any problems are encountered during the following procedure, always start over at the beginning and re-do the reference scan. Then you may add, as directed in this procedure, any type of attenuation hardware you might have reason to test.

E-2 Tools and Instruments Required

See Table E2-1

TABLE E2-1
REQUIRED TOOLS AND INSTRUMENTS

Item	Description	Part Number	Qty.
1	50-ohm dummy load, 200 watt, 30 dB attenuator - Bird Model 8322 (or equivalent).	46-317724P14	1
2	RF Test Cables Kit	46-255816G1	1

E-3 Initial Setup

1. Open the System Cabinet back door and locate the TNS module inside the upper left of the cabinet. This unit has an LCD display, Reset Button, and Disable/Enable toggle switch on the front.

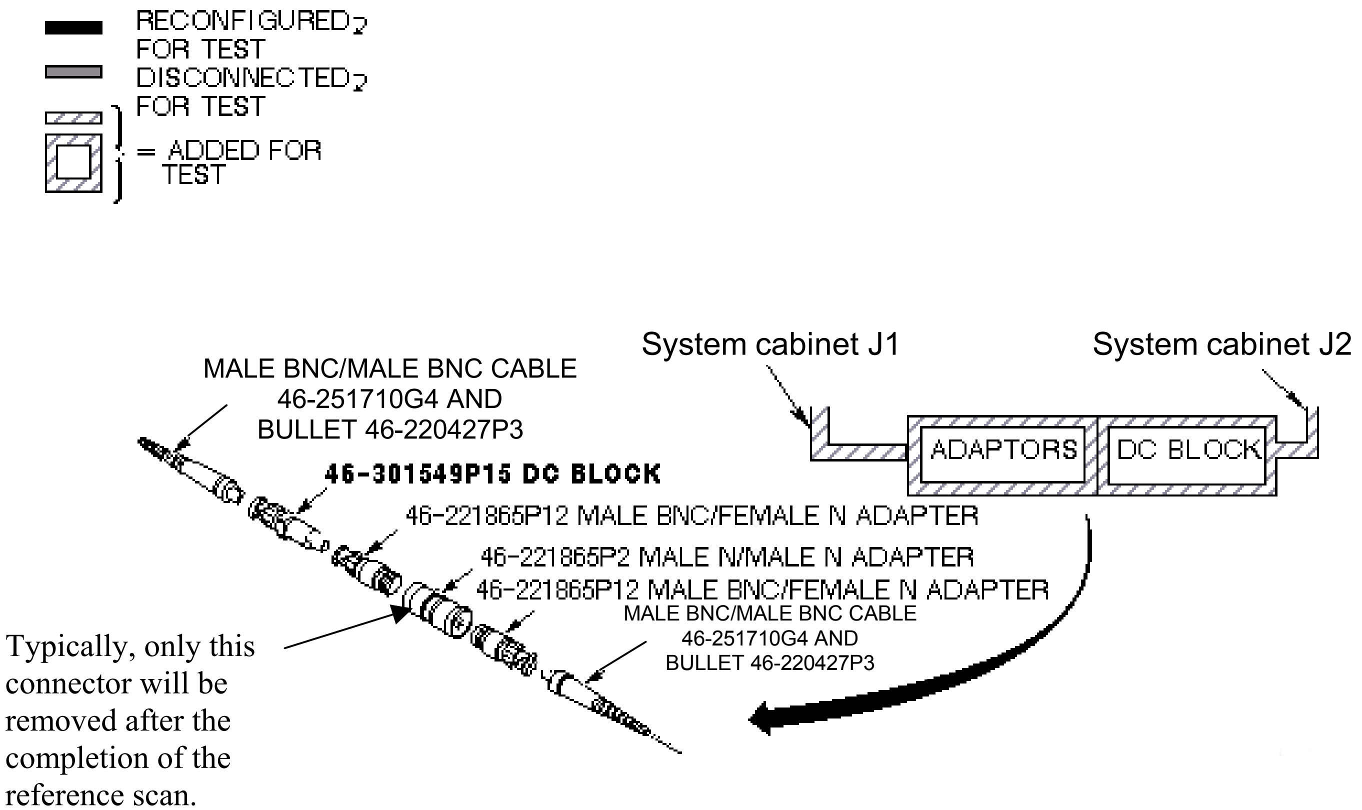
2. Locate the body TNS on the TNS module. This is one of the two shiny metal TNS boxes affixed to the main TNS module assembly that is mounted farthest from the rear of the cabinet. Multicoil systems have an additional piggyback board affixed with 3 extra TNS boxes that mounts over the top of the main TNS module assembly.

3. Bypass the body TNS (A24 A1 A1) from the circuit by disconnecting the small coax cables A24 A1 A1 J12 (signal output) and A24 A1 A1 J2 (signal input) from the TNS and connecting both together using a female-BNC to female-BNC adapter (also known as a bullet adapter 46-220427P3).

Note

Failure to bypass the body TNS out of the circuit may result no signal being received by the system. Merely disabling the TNS by moving the Enable/Disable toggle switch down to the Disable position instead of bypassing it out of the circuit may work, however, the TNS processor can override the switch.

4. Reconfigure test hardware as shown in Illustration E3-1.



CONNECTIONS FOR INITIAL AMPLITUDE SCAN
ILLUSTRATION E3-1

Note

Adapters not shown in Illustration E3-1 can be added, if necessary, from the RF Cables Kit. Usage of the inline DC Block as shown in Illustration E3-1 is mandatory.

5. Disconnect the existing cables at the Systems cabinet I/F panel J1 (RF Out) and J2 (Receiver Body Input) and set them aside.

6. Connect the assembled test cables and adapters between J1 (RF Out) and J2 (Receiver Body Input) on the System Cabinet interface.

7. At the operator workspace, prepare the system for a Dummy Load scan using the procedure, see below.

   a. Click on **[New Pt]**
      Id: **geservice**
      Name: **dummy load**
      Weight (Lb): **111**
      Set Patient Protocols to **Service**.
   b. In the Protocol field, type **o.18.1** (o=Other, 1=series number) to load the protocol.
   c. Set a landmark if necessary, then **Save Series**.
   d. With the right mouse key, select **[Research Operations]**, then select **[Display CVs]**.
      Set value of CV **calmode** to **2** (trapezoid pulse).
      (Caution here. Make sure the previous CV has been cleared before entering the next one. Look at the screen!)
      Set value of CV **p2_ramp** to **1** (1 μsec ramp time).
      Set value of CV **t2** to **50000** (50 msec tr).
      Set value of CV **pismode** to **1** (exc service).
      Set value of CV **pmode** to **1** (data collection).
      Set value of CV **daqm** to **1** (data in window).
   e. Select **[Accept]** and then select **[Research Operations]**, then select **[Download]** then select **[Manual Prescan**.

E-4 Data Collection

1. When in **[Manual Prescan]**, set **R1** to **7**, and **R2** to **14.**
2. Adjust transmit gain (**TG**) to achieve an R1 or R2 (on IP display) of approximately 98%, without going over.
3. Select **[Done]**.
4. Select **[Scan]** (Ignore the message: `MR signal too large, reduce receiver gain.`) (Note: on the LX systems tested, the scan time starts at 13 seconds, counts down to 7 seconds, then ends. This is normal and is not cause for alarm.)
5. From the MR Tools desktop, select **[Cals/checks]** and then **[Attenuation Test]**.
6. Use **[Atten Test]** tool selection to analyze data, as shown in Table E4-1.

TABLE E4-1
DATA COLLECTION

<table>
<tr><th>Output/Prompts</th><th>Input/Comments</th></tr>
<tr><td>Last run number used was: XXXX
Please enter runfile number (XXXX): ....
Please select Locked / Unlocked file (L,U) (U):..........</td><td>
<Enter>
<Enter>
(working)</td></tr>
<tr><td>|***|
****
Average Max. magnitude Across All Views = aaaaa
Average Max. magnitude Squared = bbbbb
Average RMS Across All Views = ccccc
****
Do you want to make this run the reference(Y,N)(N):......</td><td>

Y<Enter></td></tr>
<tr><td>STOP! Do not answer the next question at this point.
Continue with step 7 below.</td><td></td></tr>
</table>

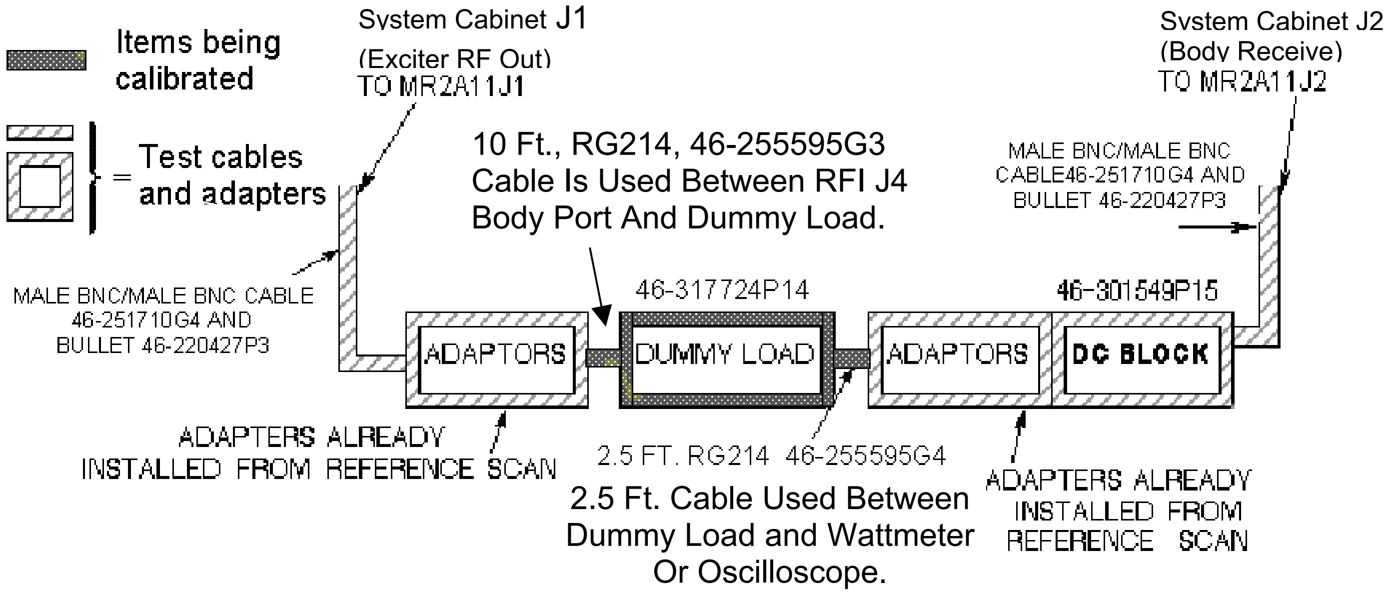
7. The next step will involve removing only the center male N to male N adapter from the test cables, setting it aside, and adding in the dummy load and cables that need to be characterized.

Note

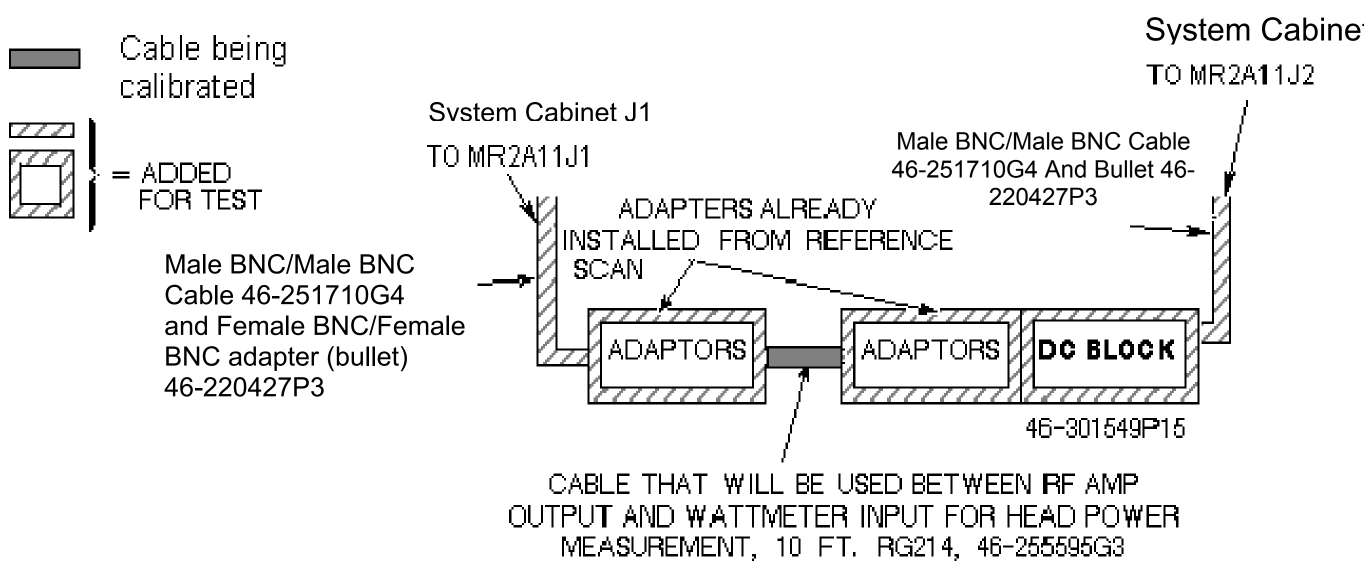
This assumes that the item to be characterized has N female connectors at it's input and output. If it has an N connector at the input and a BNC at the output or BNC connectors at both the input and output then an additional adapter(s) will be needed in order to connect it to the test cables. Adding in one or two uncharacterized adapters should not appreciably change the baseline attenuation factor. In this case, if the wattmeter is not being used, the 2.5 ft. cable (RG214, 46-255595G4) can be eliminated from the circuit. It is not needed.

8. Connect your test cables to the opposite ends of either:

- Illustration E4-1 - Dummy Load and Cables
- Illustration E4-2 - Amplifier to Wattmeter Cable.



CONNECTIONS FOR DUMMY LOAD + CABLES SCAN
ILLUSTRATION E4-1



AMP TO WATTMETER CABLE SCAN
ILLUSTRATION E4-2

Note

"Bullet" RF connector referred to in illustrations E4-1 and E4-2 is a female BNC/female BNC adapter, 46-220427P3.

9. Select the scanning icon again to activate the scanning screen.

10. Select **[Scan]**.

11. When the scan is completed, re-select the tools icon again and begin the Analysis in section E-5. It will be necessary to toggle between the Scan and Toolbelt icons if multiple passes are done.

E-5 Analysis

See Table E5-1.

TABLE E5-1
ANALYSIS

Output/Prompts	Input/Comments
`Do you want to compute Gain or Attenuation Ratio(G,A)[G]:`	**A**<**Enter**> *IMPORTANT: Answer **after** the scan is done.*
`Please do the next scan…`	
`Press <enter> key to continue, s or q to quit []:`	<**Enter**>
`Last run number used was: XXXX`	
`Please enter runfile number [XXXX]:.......................`	<**Enter**>
`Please select Locked / Unlocked file (L,U) [U]:...........`	<**Enter**> (Working)
`|***********************************|`	
`*********************************************`	
`Average Max. Magnitude Across All Views =  aaaaa`	
`Average Max. Magnitude Squared =  bbbbb`	
`Average RMS Across All Views =  ccccc`	
`Magnitude Attenuation Factor =  xxxxx`	
`Magnitude Squared Attenuation Factor =  yyyyy`	<======Record in "Value" column in Table E5-2.
`RMS Attenuation Factor =  zzzzz`	
`Do you want to make this run the reference(Y,N)(N):......`	**N**<**Enter**>
`*********************************************`	

1. Record the "Magnitude Squared Attenuation Factor" number and record it in the appropriate Value box in Table E5-2.

TABLE E5-2
ATTENUATION FACTORS

MODE	CALIBRATED HARDWARE	PART NUMBER(S)	VALUE	NOMINAL VALUES
BODY OR HEAD	**DUMMY LOAD + CABLES ATTEN FACTOR**	**46-255595G3 46-317724P14 46-255595G4**		**1000 TO 1200**
HEAD	**AMP TO WATTMETER ATTEN FACTOR**	**46-255595G3**		**1.0 TO 1.2**

E-6 Calculation of RF power

Peak voltage should be used in the calculation in order to get an accurate result. It can be converted to power using the following formula as long as certain factors are known and accounted for. The scope correction factor MUST be known. So must the *actual* total loss attributed to anything that connects the measuring device to the source. This often includes the accumulated loss associated with the dummy load and any interconnecting cables. Table E6-1 shows the calculation of power if all the attenuating devices in the measurement circuit exhibited perfect loss; that is, the devices added no more or less loss than what they were designed to provide. Table E6-2 shows the same calculation of power but accounts for the measurement-circuit loss values in deriving the true power. Note that the loss has a significant impact on the calculated power.

WARNING!

THE SCOPE CORRECTION FACTOR MUST BE KNOWN FOR THE FORMULAE SHOWN IN TABLES E6-1 AND E6-2. IF IT IS NOT KNOWN, DO NOT USE THIS METHOD. GROSSLY INACCURATE MEASUREMENTS AND POSSIBLE SYSTEM DAMAGE WILL RESULT.

$$\frac{\left(\frac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{2\text{ X Z}} \textbf{ X } \text{dummy load and cables attenuation} = \text{RF Power}$$

where “X” and “**X**” in the above formula signifies multiplication

Assume Z = 50 Ω, Vpeak = 40.0, scope correction factor = 1.00 (no loss), dummy load and cables atten. = 1000, (dummy load and cables are all ideal)

$$\frac{\left(\frac{40.0Vp}{1.00}\right)^2}{100} \textbf{ X } 1000 = 16000 \text{ Watts} = 16 \text{ kW}$$

This result assumes a ***theoretically <u>perfect</u>*** situation in which there is no loss. These situations, in common practice, rarely exist!

TABLE E6-1
RF POWER CALCULATION <u>WITH NO LOSS</u>

Now, consider the "real life" type situation in Table E6-2 in which the loss is considered:

$$\frac{\left(\frac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{2 \text{ X } Z}\ \mathbf{X}\ \text{dummy load and cables attenuation} = \text{RF Power}$$

where "X" and "**X**" in the above formula signifies multiplication

Assume Z = 50 Ω, Vpeak = 34.61, scope correction factor = 0.88, dummy load and cables atten. = 1028

$$\frac{\left(\frac{34.61Vp}{0.88}\right)^2}{100}\ \mathbf{X}\ 1028 = 15901 \text{ Watts (very close to 16kW.)}$$

Accounting for the loss resulted in an accurate answer. 15899 Watts is as close as we can hope to get to 16000 Watts without using the RF Power Measurement Kit. Note that if none of the loss had been accounted for the error could have been **3587 Watts or 22.6%.** As a result, the observer would attempt to adjust the RF power far above the 16000 Watt limit!

TABLE E6-2
RF POWER CALCULATION WITH LOSS CONSIDERED

E-7 System Restoration

1. Reconnect original cables to the System Cabinet I/F J1 and J2.
2. Remove the female-BNC to female-BNC (bullet) adapter joining A24 A1 A1 J12 and A24 A1 A1 J2 and reconnect these to the body TNS. J2 (signal input) will connect to the top of the TNS and J12 (signal output) will connect to the bottom.
3. Perform one satisfactory head or body scan.

E-8 Wattmeter Trivia

Why are measurements between a oscilloscope and a Bird Wattmeter found to be different? Repeated GEMS Education Center lab groups that performed the Dummy Load and Cables attenuation process correctly, and obtained an accurate attenuation value, were able to accurately measure RF power as well as with a 400 MHZ scope! Remember, for wattmeter measurements however, the meter reading must be multiplied by the atten factor. Leaving this little factor out is the reason why the above 'note' was made. In body mode the wattmeter is placed after the dummy load. Its atten value must be used to calculate the correct RF value. If it is NOT used, then it should be expected there will be an error of 10% or more!

In head mode, the wattmeter is placed before the dummy load. In this case only the 'cable' factor is used to correct the wattmeter reading. The 'cable' factor is the other atten test you did for the 10 foot cable only. Typically that value is between 1.0 to 1.2. Doesn't sound like much???? Take your calculator out for a moment. Take 2,000 watts (displayed reading) and multiply by an atten factor of 1.0 ; the results are obvious, 2,000 watts. Now take the same displayed 2,000 and multiply it by an atten factor of 1.2 = 2,400 watts! Hopefully this example will show the importance of the attenuation values. You just calculated a 20% error if you were wrong because you didn't do atten test on the 10 foot cable!

E-9 Scope Trivia

A basic 'rule of thumb' in the scope industry is that 'to accurately measure a high frequency signal, the bandwidth of the scope should be at least 5X the frequency being measured'. That means that for a 64 MHZ signal the scope bandwidth would have to be at least 320 MHZ. Along the same line, a 42 MHZ signal would require a bandwidth of at least 210 MHZ. This is why the 400 MHZ scope was chosen several years ago for MR use. Knowing this should alert you as to what might have to be done in order to get correct scope readings from whatever scope you carry today. Who knows what bandwidth scope you'll be working with tomorrow? Will it be of sufficient bandwidth to give accurate measurements? What can you do if it is not covered by the rule of thumb? Read on!

Experimentation with multiple scopes have shown that an error is typically somewhere between 0 and 16%. That's right, some 100 MHZ scopes have no loss! Others may have as much as 15 or 16%. Additionally, it has been found that it is not necessarily the whole scope that has this loss, but the individual channels typically exhibit differences. In other words, channel one might have a 3% loss when channel two could lose 15%. There is no fixed number for all 100 MHZ scopes!

How do you get around these losses? Well, there are two ways. First you could compare a RF waveform on your scope, side by side, with a 400 MHZ scope. Use the same cables etc. and just move the cables from scope to scope. You can then calculate the loss for <u>each</u> channel of the 100 MHZ scope vs. the 400 MHZ scope. Keep the scope intensity low for best accuracy. If and when possible, determine the losses for a 1.5T and 1.0T system. That's for 64 and 42 MHZ. Point of note: there are various losses at 42 MHZ as well as with 64 MHZ. And, typically they differ! Your scope may have a 0 to 15% rolloff at 64 MHZ and 0 to 12% at 42 MHZ. All scopes differ!

Secondly, the next time you have your scope calibrated, just attach a tag asking for the % loss of signal at 42 and 64 MHZ <u>for each channel</u>. You should also state that you want this to be done at the 5 and 2 volts/div setting. This will cover both body and head measurements. At two different re-calibration vendors we use, there are no extra charges incurred for this request.

As mentioned above, the 100 MHZ scope has been shown to be as accurate as a 400 MHZ scopes if the scope rolloff factor is known. Additionally, each channel has it's own unique loss factor.

The biggest error found in GEMS Education Center labs when measuring RF power has proven to be the FE using "typical" dummy load and cables attenuation factors. Basically there are no "typical" factors! All loads and cables are somewhat different. Several may appear to be close in value, but in most cases they are different. If you use a "typical" atten factor in your calculations, then "typically" you will be in error.

Through experimentation at the GEMS Education Center, it has been determined that if you know the attenuation factor for a particular dummy load and cables, those values hold over several years. They do NOT tend to change. This means that if you could do atten test on a dummy load and cables, and keep them as a kit, you don't have to recalibrate them every time you use them. Wow! And possibly a little cheaper than the Power Measurement kit.

If you must, or intend to use a scope to measure body RF power, use atten test on the dummy load and cables.

Knowing the correct attenuation values and the correct scope rolloff values will provide you with measurements as accurate as a 400 MHZ scope and / or the GE Power Measurements kit.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	December 11, 2000	Don Thome'	New 1.5T proprietary SRF document.
1	April 20, 2001	Don Thome'	Added reference to MONS1.EXE.
2	July 6, 2001	Don Thome'	Added reference to Power Calculator Tool throughout document. Corrected various errors.
3	January 4, 2002	Don Thome'	Added information concerning use of MONS1.EXE software.
4	March 1, 2002	Don Thome'	Corrected error in statement after Illustration 4-4 on page 11.
5	May 20, 2003	D. Hofstetter	Added section that explains how to use mons1.exe on Windows 2000.